



### Putting a 3D printer to space

"Made in Space" is in works to put a 3D printer with robotic arms into orbit to build satellites and telescopes. <http://digit.in/3DPrintARM>



### A supercomputer in space

NASA and HP want to get a supercomputer to the International Space Station, and SpaceX will take it there. <http://digit.in/SpaceXpc>



# How to be a DIY Astronomer

Looking to dive deep into the cosmos, first hand? Here's how to get started.

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**W**e will start this DIY with the assumption that anyone who wants to be an amateur astronomer has a basic understanding of astronomical concepts like our solar system, planets, stars, satellite bodies like the moon etc. Now, of course, you need to know a lot more to qualify even as an amateur but brushing up your basic celestial concepts would be a good idea.

## STEP 1: READ UP... AND READ UP SOME MORE

Hell, even the most informed and well known astronomers of our generation will admit to truly not knowing much about the big expanse and what's out there. But of what is known, there's a lot you need to familiarise yourself with. For instance, to even get started with stargazing, you need to

be familiar with the night sky and the way it's laid out. You will also need to understand concepts like the celestial sphere, how the sky moves and more. Check out a guide on how to read a star chart here: <http://digit.in/StrChrt>. There's another guide that teaches you how to measure angles in the night sky here: <http://digit.in/SkyMsr>.

## STEP 2: LOOK UP.. AND BRING YOUR SMARTPHONE ALONG

If you've tried your hand at some star maps, now would be the right time to try and locate those constellations. And there's a particularly easy tool to do that – your smartphone.

Use a star mapping app on your phone – a particularly good one being the Sky Map for Android (<http://digit.in/SkyMapApp>) – as a guide to the night sky. Use the toggles to only see things that you've understood so far, or you might be overwhelmed.

## STEP 3: CELESTIAL MINGLING

Once you've done your reading, familiarised yourself with astronomy lingo and believe that you can hold your own in a conversation about the cosmos, it's now time to go out there and get to know the community.

There are quite a few good astronomy clubs in India which welcome amateurs into their folds. The Astronomical Society of India was established in 1972 and now has close to 1,000 members. While it is a bit difficult to become a member, even as a student, since they require approval from two existing regular members, it is highly likely that you can reach out to an existing member to present your case. They maintain a list of existing members here: <http://digit.in/ASImem>.

A couple of other significant ones are:

- **Bangalore:** Association of Bangalore Amateur Astronomers.
- **Delhi:** Amateur Astronomers Association, Delhi (AAAD).
- **Pune:** Jyotirvidya Parisanstha

On social media, a couple of really helpful places to post all your queries at any level can be found here: <http://digit.in/AstrCos> and here: <http://digit.in/AstrNo1>. You can also use services like Meetup where you can find multiple enthusiast groups and even the chance to try out your hand at some actual stargazing equipment.

## STEP 4: GET YOUR OWN GEAR.. OR BUILD IT

Before you google 'best telescopes under....' keep in mind that often the best first telescope is a pair of binoculars. Once you've ticked off most things on this list: <http://digit.in/BfrTScp>, check out a detailed guide on understanding which telescope you actually need here: <http://digit.in/BuyTScp>. And if you're up for the task, you could even build your own telescope! Check out a basic guide from NASA here: <http://digit.in/NASATScp> and an advanced one on Instructables here: <http://digit.in/DobsTScp>. You'll find most of your required materials online on websites like Amazon.in.

## STEP 5: STARGAZE

Find a good nice spot with a clear sky above your head, preferably somewhere away from the blinding city lights. Do let us know about your experience at [editor@digit.in](mailto:editor@digit.in).

**Extra resources:** <http://digit.in/Astrnmy>.